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# Experimental investigation of thermoacoustic coupling using blended hydrogen–methane fuels in a low swirl burner

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ABSTRACT

In this study, experimental testing and analysis were performed to examine the combustion instability characteristics of hydrogen–methane blended fuels for a low-swirl lean premixed burner. The aim of this study is to determine the effect of hydrogen addition on combustion instability, and this is assessed by examining the flame response to a range of constant amplitude, single frequency chamber acoustic modes. Three different blends of hydrogen and methane (93% $CH_4$–7% $H_2$, 80% $CH_4$–20% $H_2$ and 70% $CH_4$–30% $H_2$ by volume) were employed as fuel at an equivalence ratio of 0.5, and with four different acoustic excitation frequencies (85, 125, 222 and 399 Hz). Planar laser induced fluorescence of the hydroxyl radical (OH-PLIF) was employed to measure the OH concentration at different phases of acoustic excitation and a Rayleigh Index was then calculated to determine the degree of thermoacoustic coupling. It was found, as has been previously reported, that the combustion characteristics are very sensitive to the fraction of hydrogen in the fuel mixture. The flame shows significant increases in flame base coupling and flame compaction with increasing hydrogen concentration for all conditions. While this effect enhances the flame response at non-resonant frequencies, it induces only minimal compaction and appears to decreases the coupling intensity at the resonant frequency.

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## 1. Introduction

In combustion systems operating with lean mixtures, one of the most critical current issues is combustion instability, i.e. large pressure oscillations that are caused by coupling between the thermo-acoustics and the fluid dynamics. Combustion instability may reduce system performance, induce vibrations, reduction the efficiency of the combustion, cause structural damage, etc. Understanding how combustion instability occurs is therefore very important in terms of applications such as gas turbine, furnaces, industrial burners, and the like. The range of physical interactions that result in combustion instability is not clearly understood and there are currently multiple open questions. This study expands on previous work with the Low Swirl Burner where the focus has been on initiation of shear layer vortices as a key step in the flame becoming unstable. This study focuses on the effect of hydrogen addition and the resulting flow structures. The relevant literature falls into several areas: acoustically forced flames, hydrogen addition to unstable flames, and general literature on topics such as flame speed.

Ghoniem et al. [1] studied lean premixed combustion stabilized behind a backward-facing step using a propane-air mixture. They showed that hydrogen addition to propane as the primary fuel improves the flame stability over the entire range of the air jet mass flow and reduces pressure oscillations.

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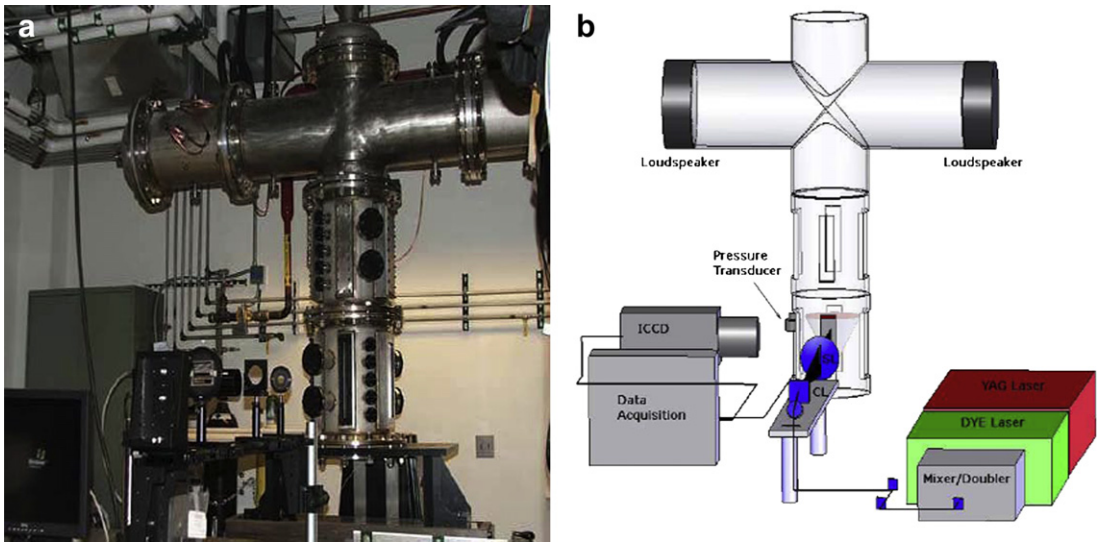


**Fig. 1 – The experimental system: (a) Photo (left), (b) Schematic (right).**

An experimental and computation study of the effects of hydrogen addition to methane on lean premixed flame extinction and response to stretch was carried out by Jackson et al. [2], but it did not directly assess nor predict combustion instability. Their experimental and numerical results indicated that increasing hydrogen in the fuel significantly increases flame speeds and thus extinction strain rates. Ilbas et al. [3] measured the laminar flame velocities of hydrogen–air and various compositions of hydrogen–methane–air mixtures at ambient temperatures for variable equivalence ratios. They found that the flame speeds increase with hydrogen content in hydrogen–methane mixtures, and thus the burning velocities also increased dramatically. These trends were expanded to elevated pressures by Hu et al. [4], which provide a way of scaling the tests in this study to more realistic conditions. Lieuwen et al. [5] investigated the impact of fuel composition on the operability of lean premixed systems. They assessed multi-species mixtures and determined that varying the hydrogen fraction had the most dramatic effect on the turbulent flame speed and ignition delay time. Unfortunately, this study did not include examination of acoustic forcing and how the flame acoustic response varies due to hydrogen fraction. Schefer et al. [6] showed that hydrogen addition expanded the stable operating envelope of a swirled flame by enabling flame anchoring. This was experimentally confirmed and elucidated with modeling by Choudhuri and Gollahalli [7]. Control of combustion instability [8] and emissions [9] has also been demonstrated through hydrogen addition.

One issue related to the measurement of heat release by quantification of OH concentration (as is often done in these studies) is the amount of OH produced by hydrogen fuel versus methane fuel. Assumptions can be made about the quasi-steady nature of the flame but validation requires a different type of measurements. Such a measurement was performed with a new diagnostic method of planar laser-induced fluorescence spectroscopy by Katoh et al. [10] to observe two types of OH radicals produced from the combustion of hydrogen and methane premixed flames. Their results show that the assumption is reasonable of behavior of hybrid fuel flames. Ciani et al. [11] investigated experimentally the stability of methane-air and hydrogen–air flames in an axisymmetric counterflow burner. Their experimental results showed that the flow field, including reactive and nonreactive cases, exhibits bistability phenomena which play an important role for the flame morphology, stability and transitions. Equivalence ratio fluctuations can also play a role in stability where an outer flow has significant interaction with the flame [12]. Work by Wicksall et al. [13] determined instantaneous velocity- field measurements experimentally in an enclosed swirl-stabilized burner operated on methane and hydrogen-enriched methane. They reported that the average and instantaneous velocity fields were affected by the addition of hydrogen to methane. Muruganandam et al. [14] developed an active control system which permits turbine engine combustors to operate safely closer to the lean-blowout limit. They also note that the system prevented lean blowout, while minimizing the pilot fuel, and therefore also minimizing the NOx. An experimental study was performed by Zhang et al. [15] on the dependence of lean blowout limits upon fuel composition for hydrogen–carbon monoxide–methane mixtures. They observed that the flames can be stabilized at lower equivalence ratios, adiabatic flame temperatures, and laminar flame speeds with increasing hydrogen in the mixture. They also note that the blow-off phenomenology changes considerably depending hydrogen levels in the mixture. Lieuwen et al. [16] reported the impact of syngas fuel composition on combustor blowout, flashback, dynamic stability, and auto-ignition in premixed combustion systems. In the current literature, other studies [17–22] about

**Table 1 – Experiment conditions (P = 1 bar).**

| Fuel (by vol.) | Equivalence ratio | Acoustic frequency (Hz) |
|---|---|---|
| 7%$H_2$–93% $CH_4$ | 0.5 | 85, 125, 222, 399 |
| 20%$H_2$–80%$CH_4$ | 0.5 | 85, 125, 222, 399 |
| 30%$H_2$–70%$CH_4$ | 0.5 | 85, 125, 222, 399 |

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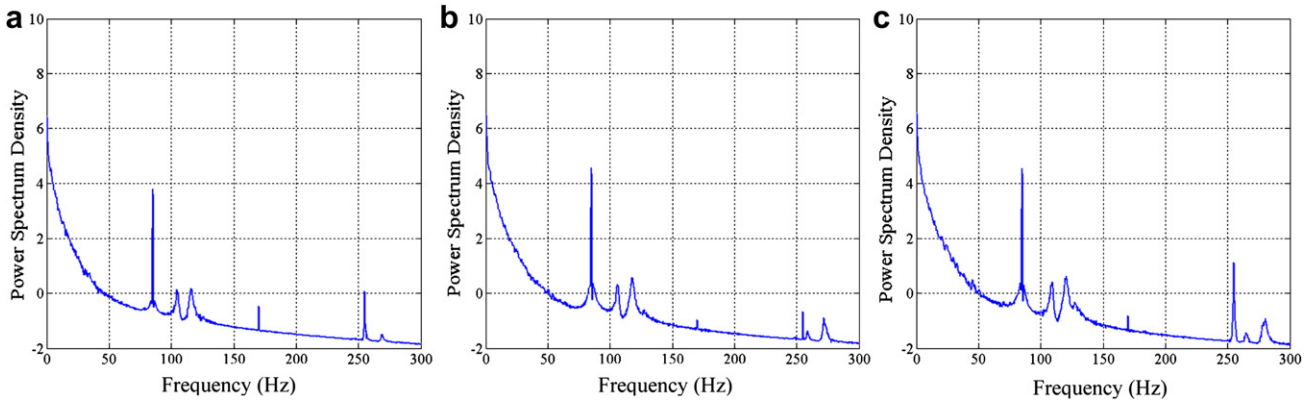


**Fig. 2 – Power spectrum for pressure ($f = 85$ Hz): (a) 7% $H_2$, (b) 20% $H_2$, and (c) 30% $H_2$.**

combustion instability were obtained only for naturally oscillating systems [23].

While the studies above show that hydrogen is a critical factor in mixed fuel flames as it dominates the flame speed when it reaches 10% to 15% of the mass flow, an open question is what effect hydrogen will have on the shear-layer tripping mechanism reported by Kang et al. [24]. This mechanism is tied to shear-layer vortices that induce changes in flame area, and thereby affect the heat release. This was documented in Yun and Ratner [25] for this burner, with resulting flame behavior that exhibits particular traits as described in the book by Lieuwen and Yang [26]. What was novel in the mechanism identified by Kang and co-workers was that the pressure amplitude (of an imposed acoustic wave) required to initiate phase-locked vortices was much smaller (0.05% of mean pressure) than what was normally seen (5% of

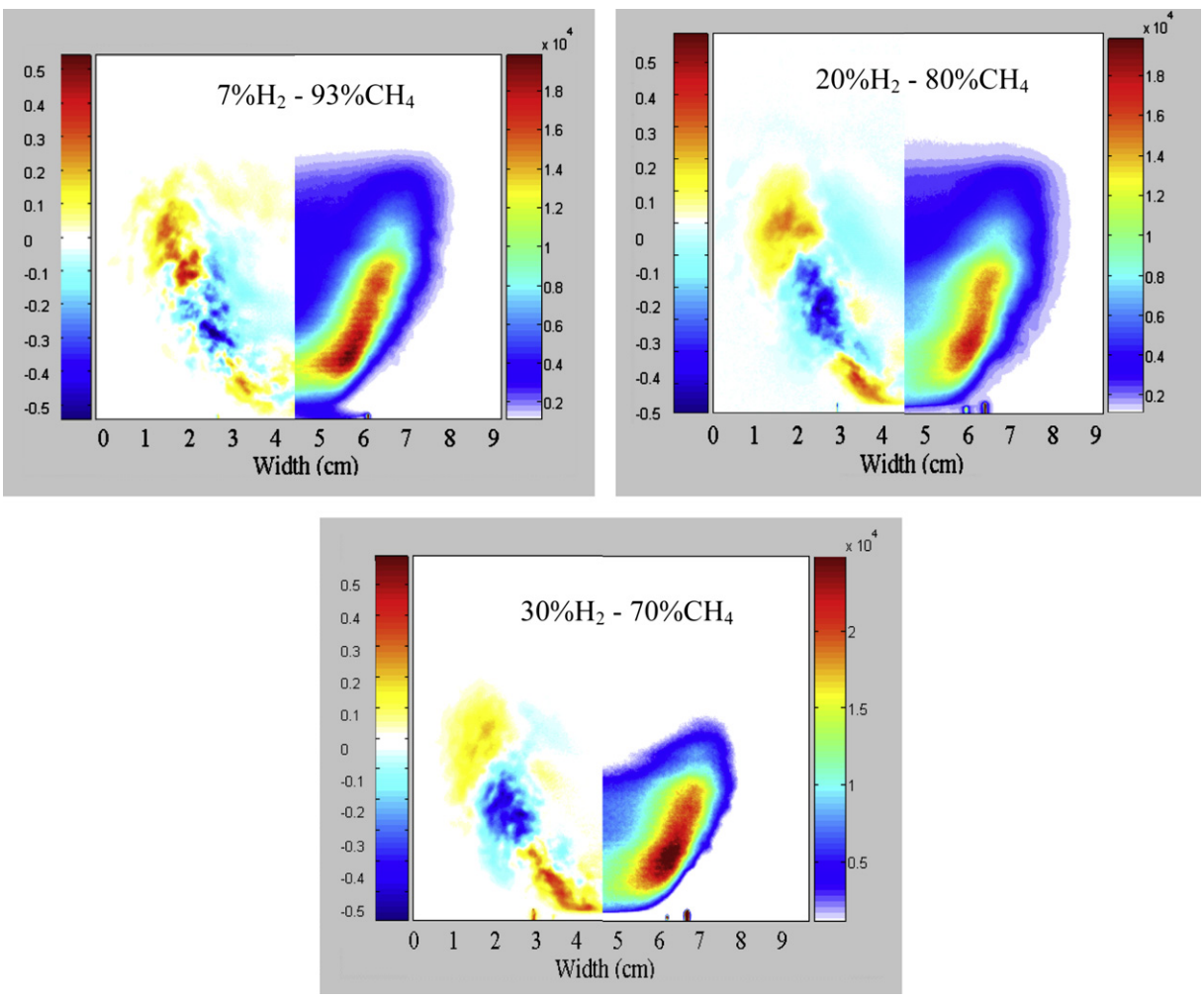


**Fig. 3 – Mean OH concentration (right) and Rayleigh Index (left) at 85 Hz acoustic forcing. ($P = 1$ atm).**

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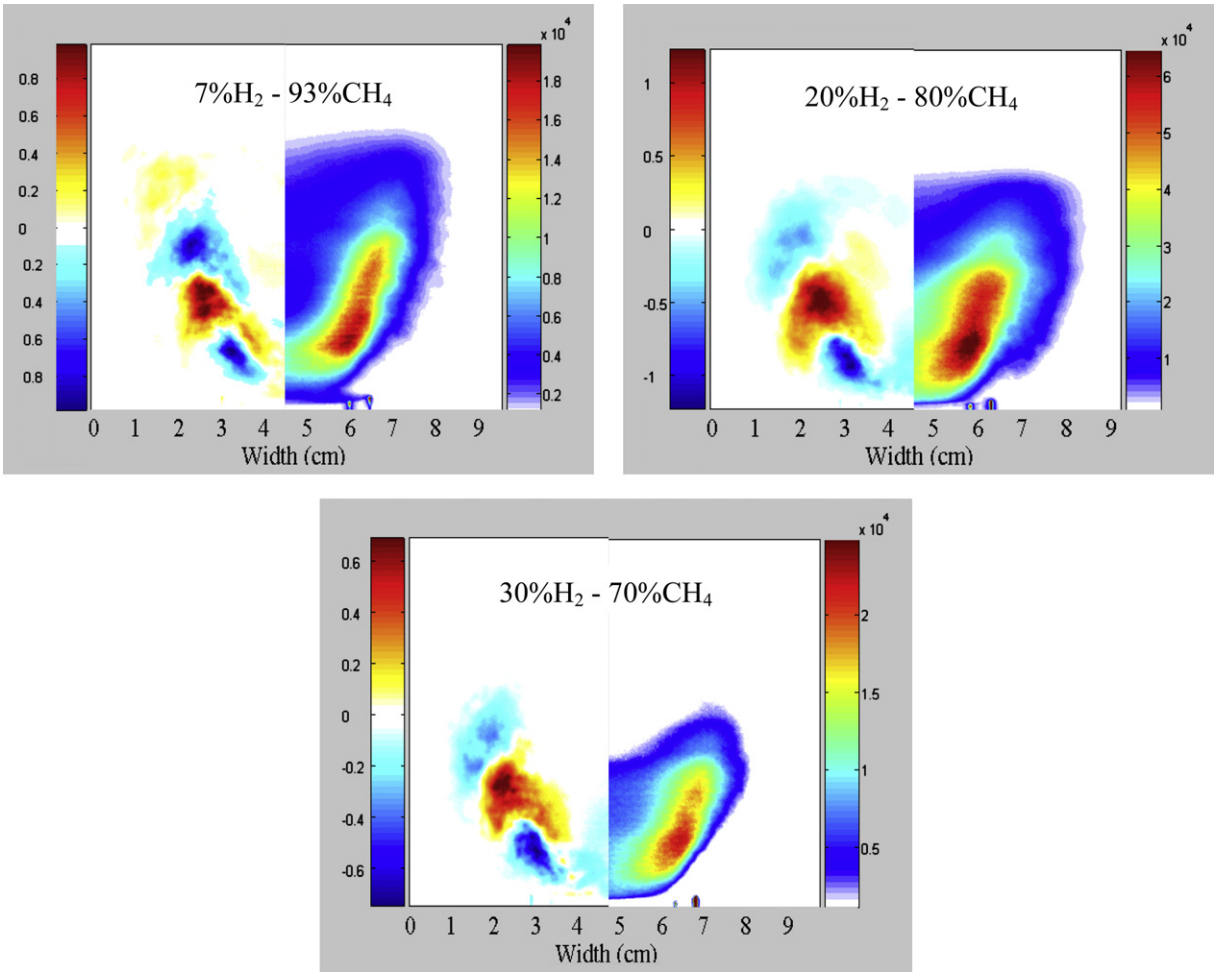


**Fig. 4 – Mean OH concentration (right) and Rayleigh Index (left) at 125 Hz acoustic forcing. ($P = 1$ atm).**

mean pressure) for standard premixed lean swirl burners [27,28]. This indicated that the vortex-tripping initiation mechanism was much more sensitive for the low swirl burner (most likely due to the absence of flame anchoring or strong flow re-circulation) than for high swirl or piloted burners. The vortex initiation mechanism is active across a fairly broad range of frequencies, from about 60 Hz to 180 Hz, and was found to be insensitive to the amplitude of the acoustic forcing (between 0.03% and 0.5% of mean pressure) [25].

The focus of the present work is to examine how the addition of hydrogen affects the vortex-tripping mechanism in the low swirl burner. The analysis methodology will mimic that employed for pure methane fueled tests and those tests should provide direct comparisons for the work here. Those efforts have also shown that this mechanism exists to at least 3 bars of chamber paper, which indicates that there is a reasonable likelihood that that this mechanism exists in commercial gas turbines with combustor pressures of 20–30 bar.

## 2. Experimental configuration

Experiments were performed using hydrogen–methane fuels in a low swirl burner, with a swirl number of 0.5 and 65 degrees of vane angle. The experimental system consists of a laser system and its associated components, the laser transmission and sheet forming optics, the burner section (burner, pre-mixer, swirler, and gas supply system), the stainless steel combustion chamber with acoustic drivers and its controller, and the ICCD-based data acquisition system. The pressure oscillations are created by four 30-cm-diameter speakers that are powered by a 2600 W power amplifier (Mackie M1400i). A photo of the experiment system and an experiment schematic are shown in Fig. 1. The combustion system for the tests presented here was operated at atmospheric pressure. The combustion air was supplied by a high-volume rotary-screw compressor while the fuels, methane and hydrogen, were supplied from separate pressurized cylinders. Combustion was started using methane and when the methane flame reached steady state, hydrogen was added to achieve the desired fuel ratio.

The premise in this experimental configuration is that the acoustic forcing is both weak and chamber-based. This means that there is minimal impact on the fuel supply system and, for the range of frequencies tested, that the flame experiences a bulk (spatially uniform) rise and then fall in pressure (versus a traveling pressure wave). For even the 399 Hz test case, there is less than a 7% variation in pressure amplitude across the flame. To ensure acoustic driving amplitude uniformity and

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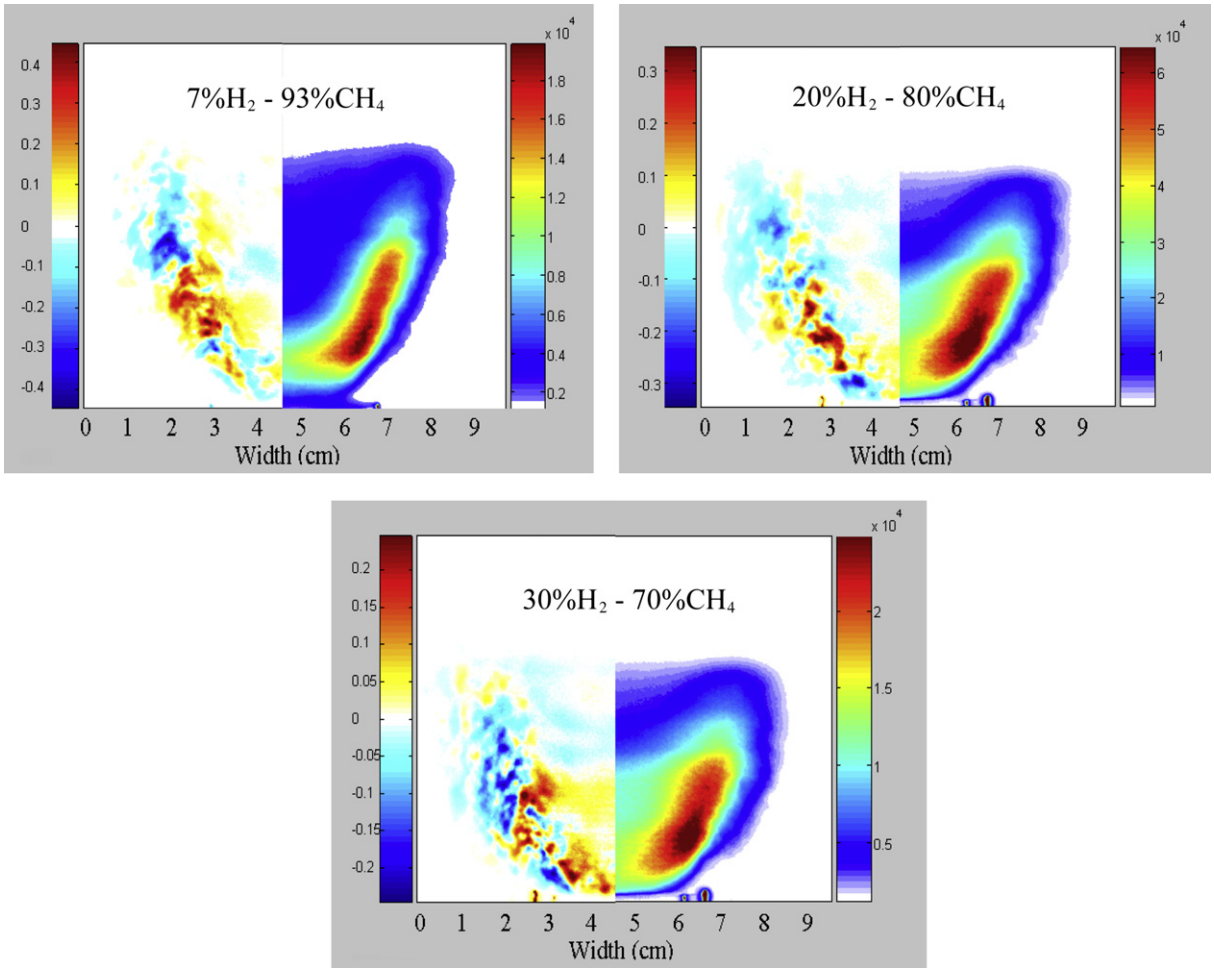


**Fig. 5 – Mean OH concentration (right) and Rayleigh Index (left) at 222 Hz acoustic forcing. ($P = 1$ atm).**

single frequency driving, a pressure sensor is located near the flame and an active control system is engaged to ensure uniformity. The acoustic amplitude was 0.05% of mean chamber pressure in all test cases. Further details regarding experimental configuration and the system can be found in refs.[23–25].

The equivalence ratio is calculated based on a method first proposed by Yu et al. [29] based on their analysis of laminar flame speed data for hydrocarbon fuels blended with hydrogen. The mole fraction of the hydrocarbon fuel, hydrogen, and air are taken as $C_f$, $C_H$, and $C_A$, respectively, with $C_f + C_H + C_A = 1$. Assuming the $H_2$ is fully oxidized, $C_H/(C_H/C_A)$ is the amount of air required, which is equal to $C_H/(C_H/C_A)_{st}$, where $(C_H/C_A)_{st} = 0.418$ is the stoichiometric hydrogen to air molar ratio. If the remaining air is used to oxidize the hydrocarbon fuel, an effective fuel/air equivalence ratio $\phi$ can be defined as

$$\phi = \frac{C_F/\left[C_A - C_H/(C_H/C_A)_{st}\right]}{(C_F/C_A)_{st}} \quad (1)$$

where $(C_F/C_A)_{st}$ is the stoichiometric fuel to air molar ratio. The benefit of this method is that it provides simple comparisons when the hydrogen fuel fraction is small. The drawback is that at higher percentages the equivalence ratio calculated by this method does not account for the hydrogen combustion, and would not indicate the higher flame temperatures that would occur in those flames. Experimental conditions used in the study are given in Table 1. As shown in the table, the fuel-air mixtures are all tested at the same equivalence ratio, with the known peak temperature change, and are tested at the same four forcing frequencies: 85, 125, 222, and 399 Hz.

The images of OH concentration are collected randomly during the acoustic cycle and then collected into 10 degree phase bins in post-processing (with 10 images per bin being typical). These images are then averaged to create a single image for each portion of phase. Since the pressure is uniform over the entire image, the 36 averaged images can be directly used to create Rayleigh Index images. Since the selected acoustic frequency is much higher than everything but very low frequency noise, even 10 image averages tend to be fairly low in noise.

## 3. Results and discussion

Fig. 2 shows the pressure oscillations for the different fuel blends, plotted as the power density, here taken as $Log_{10}$(-Pressure). The amplitude for the 85 Hz driven frequency varies even though the active control tries to maintain constant amplitude. This is due to the inherent difficult in maintaining

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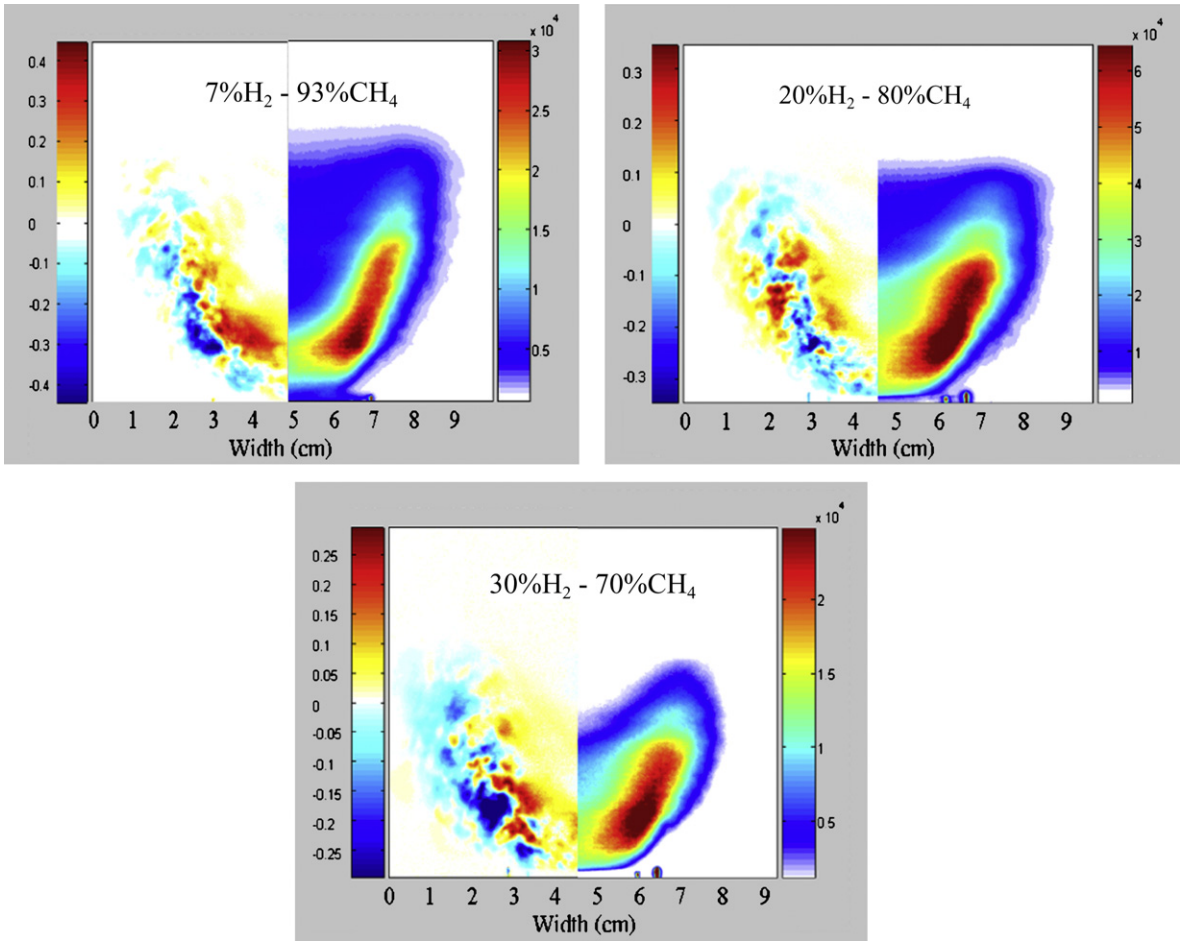


**Fig. 6 – Mean OH concentration (right) and Rayleigh Index (left) at 399 Hz acoustic forcing. ($P = 1$ atm).**

a low amplitude signal while canceling other noise and harmonics. The effect this has on the experiments should be minor die to the amplitude insensitivity described in section 1. The 125 Hz natural frequency is evident in all three cases, but is much weaker than the driven frequency. The third harmonic at 255 Hz is seen to vary significantly, but since the active control system tries to cancel out the harmonics, these differences are likely due to some combination of flame and controller effects. Since they are always significantly weaker than the main peak, they are also, like the natural frequency, not considered in the analysis.

Rayleigh index has long been considered as an important indicator of thermoacoustic coupling. If the Rayleigh index is positive, pressure oscillation and heat release oscillation are in phase and tend to grow. If the Rayleigh index is negative, the oscillation tends to decay. In this study, this index is examined as a major system parameter. The simplified normalized Rayleigh index [30,31] is defined as

$$R_f = \int_0^1 \frac{p'q'}{p_{rms}\overline{q}} d\xi \tag{2}$$

where p' is the pressure oscillation, $p_{rms}$ is the root mean square of the pressure oscillation for each set of images, q' is the oscillation of the OH intensity (heat release) of each pixel and $\overline{q}$ is the averaged total OH intensity. A key aspect of this analysis is that the changes in the OH concentration are small, and hence these small changes correspond directly to changes in the heat release rate. This aspect is also why averaged images (where random flow/flame fluctuations have been smoothed out) are necessary. Figs. 3–6 show the Rayleigh index and averaged OH-PLIF images for each of the fuel conditions for each of the acoustic forcing frequencies. All data is from the right half of the flame (where the laser sheet enters the field of view) with the averaged OH images mirrored to the left side to simplify comparison. Immediately evident for the 85 Hz case (Fig. 3) are the alternating Rayleigh index values at the boundary of the flame zone. As in the pure methane case at 85 Hz, the Rayleigh index has positive and negative zones; with the positive values (red and yellow) indicating the regions where energy is fed back to the acoustic wave while the blue-colored negative zones show where the oscillation is damped.

Several aspects are immediately evident. The zones become more intense and coherent with increasing hydrogen concentration, and the flame can be seen to get slightly shorter and flatter at the same time. This is expected as hydrogen addition will increase the flame temperature and create a more intense zone of combustion at the flame base. The flame will burn out quicker, which is why it will be shorter, and can exist in higher velocity flow, which explains

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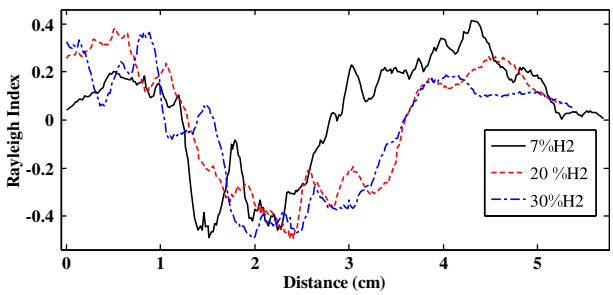


**Fig. 7 – Rayleigh Index along the flame boundary for different hydrogen percentage at 85 Hz. ($P = 1$ atm).**

the widening. What is interesting is that the Rayleigh Index for the 7% hydrogen case is more intense high in the flame and move lower with increasing hydrogen concentration. 85 Hz is in the range where shear-layer tripping occurs, but is weak, and this indicates the increasing the hydrogen fuel fraction make the flame more responsive to the acoustic forcing.

Fig. 4 shows the flame response of the blended fuels at 125 Hz acoustic forcing. This case shows many of the same characteristics as the 85 Hz tests above. The zonal structure of the Rayleigh index is again present and again intensifies with increasing hydrogen fraction. Since 125 Hz is the natural mode of the system, the zones are more clearly defined even with little or no hydrogen present. The flame, as would be expected, appears more compact at all conditions. One interesting aspect is that the intensity appears to change little and might even be getting weaker at higher hydrogen concentrations. This could be explained from the perspective that the flame is already highly coherent and phase-locked at 125 Hz and the addition of hydrogen only weakens the phase locking.

The thermoacoustic coupling (the Rayleigh index) and averaged OH concentration corresponding to 222 Hz forcing are shown in Fig. 5. This frequency is beyond the flame coupling range, and the result is that the shear layer shows little coherence and no structures related to the acoustic forcing. Hydrogen addition appears to push the positive coupling towards the base of the flame (as would be expected with hydrogen's higher flame speed), but even this effect is quite weak. These flames also show minor compaction with increasing hydrogen fraction. That implies that the greater compaction seen at 85 Hz is primarily due to enhancement of the large scale flow structures and Rayleigh index zones which then are able to respond to the hydrogen addition,

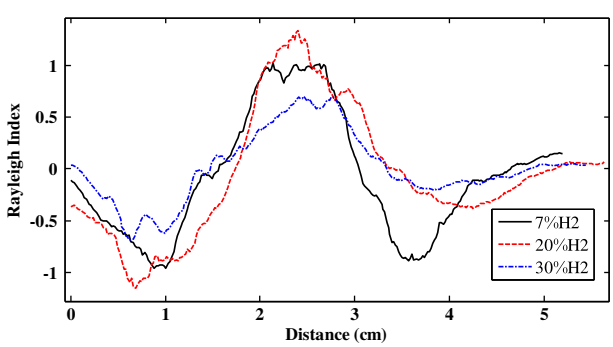


**Fig. 8 – Rayleigh Index along the flame boundary for different hydrogen percentage at 125 Hz. ($P = 1$ atm).**

increase burning intensity or flame folding, and consuming the fuel in a shorter distance (resulting in flame compaction). The final set of data, at 399 Hz, is shown in Fig. 6. The Rayleigh index fields are similar to that observed at 222 Hz.

Fig. 7 shows the variation of Rayleigh index along the flame boundary for the different hydrogen blends for 85 Hz forcing and Fig. 8 does the same for 125 Hz. The flame boundary occurs on the outer edge of the high intensity averaged OH countor. Previous work has also matched this to PIV images for this burner and the two regions match. Off-course, this is a turbulent flame and the flame zone moves significantly and the mean location is the one employed here. For 85 hz, increasing the hydrogen fraction appears to increase flame base response and appears to help induce greater flame-acoustic coupling. It appears that 20% hydrogen is sufficient to induce this close coupling and increasing the fraction to 30% has no visible impact.

Conversely, at 125 Hz it appears that increasing the hydrogen fraction decreases the peak coupling all the way to 30% hydrogen. Since the flame is already closely coupled to the forcing frequency at 125 Hz, it appears that hydrogen addition actual competes with the acoustic coupling and forces a weakening of coupling intensity.

While the trends visible in Fig. 8 are interesting, a key question is what aspects of these behaviors are beyond the expected error/uncertainty range. Rayleigh Index uncertainty for strong signal regions is approximately 10% of local intensity. Regions that have more flame intermittence often have 50% to 100% excursions. The relative error in the OH PLIF images (for regions with minimal flow-induced variability) is typically 5% or less [23–25], so the primary cause of the uncertainty/variability here is due to flow turbulence/flame displacement and the number of images over which statistical averages can be constructed. How this impacts the assessment and comparison of the current data between cases is that for differences of less than 20% (in high intensity regions), the variation is as likely to be due to statistical noise as to real physical phenomena.

There are some trends that are well beyond the error/uncertainty limits. The flame becomes more compact when hydrogen is added due to the increased flame speed. The degree of compaction varies and the data is not clear enough to show direct correlation with increasing hydrogen fraction. Unexpectedly, the degree of compaction appears to be dependent on acoustic forcing and the degree of flame coupling. Stronger coupling (at 125 Hz) appears to result in less compaction, most likely because the existing, acoustically-induced, flame coherence results in a ~20% shorter flame at even 7% hydrogen fraction. The other cases experience approximately a 30% reduction in length, with all cases producing a flame of ~5 cm in length for the 30% hydrogen fuel is employed. This is also related to the more surprising result that the shear layer coupling intensity decreases for the higher hydrogen concentration cases at 125 Hz driving. This is probably due to de-tuning of the interaction due to the increase in flame speed favoring a higher frequency interaction.

Another interesting aspect is that increasing $H_2$ seems to drive the positive coupling spatially lower in the flame. This is most likely due to the $H_2$ portion of the flame being more sensitive/reactive, and increasing the amount of $H_2$ make it easier for the (otherwise stable) flame base to respond to acoustic forcing. This implies that there is a transition somewhere in the

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20% to 30% $H_2$ range that makes the LSB flame base responsive to forcing (and probably makes it more unstable).

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